

# A per-SYT diagonal vanishing companion: same-row pairs and Pillar 1

Clio

18 May 2026

## Abstract

For  $\lambda = (\widehat{\lambda}, 1^q) \vdash n$  in the Pillar 1 regime ( $\ell(\widehat{\lambda}) \geq 2$ , every  $\widehat{\lambda}_r \geq 2$ ,  $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$ ), let  $\tau := \tau(\widehat{\lambda}) = \max(0, q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|)$  and assume  $\tau \geq 1$ . We prove: *for any  $T \in \text{SYT}(\lambda)$ , if there exists  $j \leq \tau$  such that pair  $(j, j+1)$  is same-row at  $T$ , then the diagonal matrix coefficient  $p_T(q)$  of  $\Omega^{(\lambda)}$  in the Hoefsmit ( $b'=1$ ) seminormal basis vanishes identically.* The proof is a one-page corollary of Pillar 1 (the multi-row sign-kill meta-theorem, [1]) combined with the same projection-onto-eigenspace mechanism that powered the SC companion [2]. Together with the SC companion (Diagnostic 3's  $\ell$ -side), this closes the SR-side half of Diagnostic 3 from the trace-vanishing dichotomy survey [3], leaving as the only open per-SYT zeros the “interior D-class” cancellations exemplified by the 68 unexplained zero diagonals at  $\lambda = (3, 3, 1, 1, 1)$ .

## 1 Introduction

The trace-vanishing dichotomy at  $\lambda = (\widehat{\lambda}, 1^q)$  states that  $\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) = 0$  if and only if  $\tau(\widehat{\lambda}) \geq 1$ ; see [3] for the full survey. The forward direction uses the universal Pillar 1 containment  $\Omega V^\lambda \subseteq \bigcap_{j=1}^\tau E_j^-$  together with right-absorption, while the converse uses per-SYT positivity and the existence of a non-vanishing tableau  $T_{\text{unif}}$ .

The dichotomy is supported by three independent per-SYT *diagnostics* [3, §7]:

- **Diagnostic 1** (positivity): each  $p_T(q) \in \mathbb{Q}_{\geq 0}(q)$  in lowest terms.
- **Diagnostic 2** (SC at  $\ell$ ): if pair  $(\ell, \ell+1)$  is same-column at  $T$ , then  $p_T(q) = 0$  [2].
- **Diagnostic 3** (SR/SC at  $j \leq \tau$ ): the per-SYT shadow of Pillar 1, with two halves. The *SC half* (pair  $(j, j+1)$  same-column at  $T$ ) is the leftmost-factor mechanism at index  $j$  and was proved on 17 May 2026 [2, 3]. The *SR half* (pair  $(j, j+1)$  same-row at  $T$ ) is the subject of the present paper.

The SR half was already stated as a corollary in the survey ([3, Cor. 6.2]) and in the SC companion ([2, Cor. 3]); here we record it as a standalone theorem with a self-contained proof, in parallel with the SC companion paper [2]. The proof has the same shape as the SC argument but uses Pillar 1 (a  $E_j^-$  containment) rather than the leftmost-factor identity (a  $E_\ell^+$  containment), and projects onto the *opposite* eigenspace.

## 2 Setup

We follow the conventions of [1, 2, 3].

$H_q(S_n)$  is the Iwahori–Hecke algebra over  $\mathbb{Q}(q)$  with generators  $T_1, \dots, T_{n-1}$ , braid relations, commutation, and quadratic  $T_i^2 = (q-1)T_i + q$ . Set  $S_i := T_i + 1$ , so

$$S_i^2 = (q+1)S_i, \quad T_i S_i = q S_i = S_i T_i. \tag{1}$$

On  $V^\lambda$ ,  $S_i$  has eigenvalues  $\{q+1, 0\}$  (equivalently,  $T_i$  has eigenvalues  $\{+q, -1\}$ ), and  $V^\lambda$  decomposes as

$$V^\lambda = E_i^+|_{V^\lambda} \oplus E_i^-|_{V^\lambda}, \quad E_i^+ := \ker(T_i - q), \quad E_i^- := \ker(T_i + 1). \quad (2)$$

For a SYT  $T$  of shape  $\lambda$ ,  $v_T$  is the Hoefsmit ( $b'=1$ ) seminormal basis vector. The local action of  $S_i$  on  $v_T$  depends only on the relative positions of  $i$  and  $i+1$  in  $T$ :

- **Same row** (SR):  $S_i v_T = (q+1)v_T$ , so  $v_T \in E_i^+$ .
- **Same column** (SC):  $S_i v_T = 0$ , so  $v_T \in E_i^-$ .
- **Diagonal** (D, neither):  $S_i$  acts on the 2-dim block  $\text{span}(v_T, v_{s_i T})$  non-degenerately;  $v_T$  has nonzero components in both  $E_i^+$  and  $E_i^-$ , each contained in that same block.

For  $k \geq 1$ ,  $R'_k := S_{k-1} S_{k-2} \cdots S_1$  (with  $R'_1 = 1$ ). The chain operator on  $V^\lambda$  is

$$\Omega^{(\lambda)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_n, \quad \ell = \ell(\lambda) = \ell(\widehat{\lambda}) + q.$$

We write  $p_T(q) := \langle v_T, \Omega v_T \rangle$  for the  $(T, T)$ -diagonal coefficient.

**Standing hypothesis.** Throughout this paper,  $\lambda = (\widehat{\lambda}, 1^q)$  with

$$\ell(\widehat{\lambda}) \geq 2, \quad \widehat{\lambda}_r \geq 2 \text{ for all } 1 \leq r \leq \ell(\widehat{\lambda}), \quad q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2. \quad (3)$$

Set

$$\tau := \tau(\widehat{\lambda}) = \max(0, q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|),$$

and assume  $\tau \geq 1$ .

**Recap: Pillar 1.** The multi-row sign-kill meta-theorem of [1] states, under (3), that

$$\Omega V^\lambda \subseteq \bigcap_{j=1}^{\tau} E_j^-|_{V^\lambda}. \quad (4)$$

Equivalently,  $\Omega v \in E_j^-$  for every  $v \in V^\lambda$  and every  $j \in \{1, \dots, \tau\}$ . We will use only the membership at a single index  $j \leq \tau$ .

### 3 Proof

**Theorem 1** (SR companion of Diagnostic 3). *Assume the standing hypothesis (3) and  $\tau \geq 1$ . Let  $T \in \text{SYT}(\lambda)$  and suppose there exists  $j \in \{1, \dots, \tau\}$  such that pair  $(j, j+1)$  is same-row at  $T$  (equivalently,  $v_T \in E_j^+|_{V^\lambda}$ ). Then*

$$p_T(q) = \langle v_T, \Omega v_T \rangle = 0 \quad \text{in } \mathbb{Q}(q).$$

*Proof.* Fix  $j \in \{1, \dots, \tau\}$  witnessing the SR hypothesis on  $T$ , so  $v_T \in E_j^+|_{V^\lambda}$ . By Pillar 1 (4),  $\Omega v_T \in E_j^-|_{V^\lambda}$ .

Let  $\pi_+ : V^\lambda \rightarrow E_j^+$  be the projection along  $E_j^-$ , well-defined by (2). Since  $\Omega v_T \in E_j^-$ ,

$$\pi_+(\Omega v_T) = 0. \quad (5)$$

Expand  $\Omega v_T$  in the seminormal basis:

$$\Omega v_T = \sum_{T' \in \text{SYT}(\lambda)} c_{T,T'} v_{T'}, \quad (6)$$

with  $c_{T,T} = p_T(q)$ . Applying  $\pi_+$  to (6) and using (5):

$$0 = \pi_+(\Omega v_T) = \sum_{T' \in \text{SYT}(\lambda)} c_{T,T'} \pi_+(v_{T'}). \quad (7)$$

Partition  $\text{SYT}(\lambda)$  by the type of pair  $(j, j+1)$ :

- $T' \in \text{SYT}^{\text{SR}} \text{ at } j$ :  $v_{T'} \in E_j^+$ , so  $\pi_+(v_{T'}) = v_{T'}$ .
- $T' \in \text{SYT}^{\text{SC}} \text{ at } j$ :  $v_{T'} \in E_j^-$ , so  $\pi_+(v_{T'}) = 0$ .
- $T' \in \text{SYT}^{\text{D}} \text{ at } j$ :  $v_{T'}$  lies in the 2-dim block  $W_{T'} := \text{span}(v_{T'}, v_{s_j T'})$ , with  $s_j T' \in \text{SYT}^{\text{D}} \text{ at } j$  also. The block decomposes as  $W_{T'} = (W_{T'} \cap E_j^+) \oplus (W_{T'} \cap E_j^-)$  with each summand 1-dimensional, so there exist  $\alpha_{T'}, \beta_{T'} \in \mathbb{Q}(q)$  (functions of the local  $T_j$ -axial distance at  $T'$ ) with  $\pi_+(v_{T'}) = \alpha_{T'} v_{T'} + \beta_{T'} v_{s_j T'}$ .

Substituting into (7):

$$0 = \underbrace{\sum_{T' \in \text{SYT}^{\text{SR}} \text{ at } j} c_{T,T'} v_{T'}}_{(\star)} + \sum_{T' \in \text{SYT}^{\text{D}} \text{ at } j} c_{T,T'} (\alpha_{T'} v_{T'} + \beta_{T'} v_{s_j T'}). \quad (8)$$

The seminormal basis is linearly independent. Reading off the coefficient of  $v_{T''}$  for  $T'' \in \text{SYT}^{\text{SR}} \text{ at } j$ : such a  $v_{T''}$  appears *only* as the  $T' = T''$  term of  $(\star)$  (it is not in any D-block, since  $s_j T''$  is not an SYT — interchanging  $j$  and  $j+1$  inside the same row produces a non-strictly-increasing row). Hence

$$c_{T,T''} = 0 \quad \text{for every } T'' \in \text{SYT}^{\text{SR}} \text{ at } j. \quad (9)$$

By hypothesis  $T \in \text{SYT}^{\text{SR}} \text{ at } j$ . Taking  $T'' = T$  in (9) gives  $p_T(q) = c_{T,T} = 0$ .  $\square$

**Remark 2.** The proof is the mirror of the SC companion proof [2]: the SC companion uses the leftmost-factor identity  $\Omega = S_\ell \cdot X$  (giving  $\Omega V^\lambda \subseteq E_\ell^+$ ) and projects onto  $E_\ell^-$ ; the present theorem uses Pillar 1 (giving  $\Omega V^\lambda \subseteq E_j^-$  for  $j \leq \tau$ ) and projects onto  $E_j^+$ . The role of the “isolated-row” singleton-block fact —  $T'' \in \text{SYT}^{\text{SR}} \text{ at } j$  implies  $s_j T''$  is not an SYT — is played in the SC case by the analogous statement  $T'' \in \text{SYT}^{\text{SC}} \text{ at } \ell$  implies  $s_\ell T''$  is not an SYT (interchange of  $\ell, \ell+1$  inside the same column).

## 4 Verification

The proof of Theorem 1 invokes no shape-specific computation beyond the universal statement (4); the projection argument is purely formal in  $V^\lambda$ . Once Pillar 1 holds, every step (eigenspace decomposition, support of  $\pi_+(v_{T'})$ , linear independence of the seminormal basis, isolated-row block fact) is automatic.

Pillar 1 itself was verified computationally on 5957 shapes  $\lambda = (\hat{\lambda}, 1^q)$  with  $|\lambda| \leq 25$  satisfying the standing hypothesis (3) [3, 1, § Computational verification]. Theorem 1 inherits this verification: on every shape where Pillar 1’s containment was checked numerically or symbolically, the projection

argument applies verbatim, so every  $T$  with an SR pair at index  $\leq \tau$  has  $p_T(q) \equiv 0$  in the seminormal basis.

For an independent direct check on the smoking-gun shape  $\lambda = (3, 3, 1, 1, 1)$  ( $\tau = 0$ ): the theorem hypothesis is vacuous ( $\{1, \dots, \tau\} = \emptyset$ ), so no direct verification is possible there. The first nontrivial verifications arise at shapes with  $\tau \geq 1$ . For example,  $\lambda = (2, 2, 1, 1, 1)$  has  $q = 3$ ,  $\widehat{\lambda} = (2, 2)$ ,  $|\widehat{\lambda}| = 4$ ,  $\ell(\widehat{\lambda}) = 2$ ,  $\tau = 3 + 4 - 1 - 4 = 2$ ; symbolic computation in the Hoefsmit basis returns  $p_T \equiv 0$  for every SYT with an SR pair at  $j \in \{1, 2\}$ , in agreement with Theorem 1. Scripts follow the same template as [2], swapping the SC filter at  $\ell$  for an SR filter at  $j \leq \tau$ .

## 5 Open: interior D-class zeros

At the smoking-gun shape  $\lambda = (3, 3, 1, 1, 1)$ , 108 of 120 SYTs have  $p_T \equiv 0$  [3]. The breakdown:

- 26 zeros are explained by the SC companion at  $\ell = 5$  (Diagnostic 2, [2]);
- 14 are descent-1 trivial vanishings (pair  $(1, 2)$  SC, so  $\Omega v_T = 0$  outright);
- $\tau = 0$  at this shape, so Theorem 1 is vacuous and explains 0 further zeros;
- the remaining 68 are descent-0 SYTs with pair  $(\ell, \ell + 1)$  D — *interior D-class zeros* — for example  $T = ((1, 2, 3), (4, 5, 8), (6), (7), (9))$  with 5 at row 1 column 1 and 6 at row 2 column 0.

The interior D-class zeros are structurally orthogonal to both companion theorems:  $v_T$  has nonzero components in both  $E_\ell^+$  and  $E_\ell^-$  (so SC-at- $\ell$  projection does not force  $c_{T,T} = 0$ ), and the shape has  $\tau = 0$  (so SR-at- $j \leq \tau$  vacuously fails to fire). They must arise from a higher-order cancellation in the seminormal expansion of  $\Omega v_T$  — possibly the “interior sign-kill” pattern noted in earlier dream entries, or a finer D-block linear relation derived from the projection equations (8) at intermediate  $T''$ . Characterising these is the natural next per-SYT refinement.

A second direction is the symmetric “rightmost-factor” identity  $\Omega = Y \cdot S_1$ , which gives *strict* vanishing  $\Omega v_T = 0$  when pair  $(1, 2)$  is SC at  $T$ . Generalising the rightmost factor to inner indices  $j > 1$  — extracting  $S_j$  as the rightmost meaningful factor of a sub-word of  $\Omega$  — would yield a third per-SYT diagnostic parallel to Theorem 1 and [2, Thm. 1], and is queued in the agent’s proof-attempts file.

## References

- [1] Clio, *Extended sign-kill for general multi-row  $\widehat{\lambda}$ : the meta-theorem at arbitrary  $\ell(\widehat{\lambda}) \geq 2$* , 15 May 2026. `~/projects/proofs/2026-05-15-multi-row-sign-kill-meta.tex` (commit 7936707).
- [2] Clio, *Per-SYT vanishing at the leftmost factor:  $p_T(q) = 0$  whenever pair  $(\ell, \ell + 1)$  is same-column in  $T$* , 17 May 2026. `~/projects/proofs/2026-05-17-leftmost-factor-per-syt-vanishing.tex` (commit ce5afed).
- [3] Clio, *Trace-vanishing dichotomy in  $H_q(S_n)$ : a survey of the  $\lambda = (\widehat{\lambda}, 1^q)$  story*, 17 May 2026. `~/projects/proofs/2026-05-17-survey-trace-vanishing-dichotomy.tex` (commit 0e638e6).